UV Finiteness and Renormalization Group Flow of Curvature-Memory Scalar-Tensor Gravity

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2026

Abstract

We establish the ultraviolet structure of Curvature-Memory Scalar-Tensor Gravity (CMSTG) at one and two loops in the resummed propagator. The theory is defined by the canonical action of Paper II [16] together with a non-local memory kernel $e^{-k^2/{k_m}^2}$ postulated as a defining feature of the scalar sector and characterised by a single physical scale k_m. We do not derive the kernel from a more fundamental theory; we treat it as an input that specifies the model, in the same sense that the Wilsonian cutoff specifies an effective field theory. With this input, four numerical simulation programmes (SIM102, SIM104, SIM105, SIM106) establish the following. **(i)** The bare one-loop Ψ self-energy is quartically divergent; the memory kernel regulates it to the finite result $\Sigma(0) = {\Lambda_0}^2 {k_m}^4/(64\pi^2)$, confirmed numerically at better than 0.22% across two orders of magnitude in k_m. **(ii)** All four 1PI diagrams at one loop are finite in the resummed theory: the Ψ self-energy (regulated by the kernel), the graviton wavefunction renormalization (protected by the diffeomorphism Ward identity $\Pi_{hh}(0) = 0$), the vertex correction, and the two-loop Ψ self-energy. The three contributions form a hierarchical expansion with magnitudes 1, 2.4×10^{-2}, and 3.4×10^{-6} at the naturalness threshold $k_m^{\mathrm{nat}} = 9.15\,\mathrm{Mpc}^{-1}$. **(iii)** The renormalization-group flow of Λ_0 at fixed k_m is negative at all scales: $\beta(\Lambda_0) = -{\Lambda_0}^3 {k_m}^2/(16\pi^2 m_0^2)$. The running coupling has a UV fixed point at $\Lambda_0 = 0$ (the GR limit) and no infrared Landau pole. We refer to this as a "Λ_0 fixed point" rather than asymptotic freedom in the strict sense, because k_m is a fixed input of the theory rather than a coupling that runs. **(iv)** At two loops, the graviton sector is UV-finite via three independent mechanisms: the Ward identity, Dyson resummation of the Ψ propagator, and the finiteness of $dI/dp^2|_0$ in $d = 4$. The graviton wavefunction renormalization $\Delta Z_h/Z_h \approx 1.4 \times 10^{-6}$ is negligible. These results establish that CMSTG is UV-finite to two loops in the resummed propagator, with an explicit RG trajectory connecting the observed $\Lambda_{0\mathrm{obs}} = 0.003\, M_{\mathrm{Pl}}^{-2}$ to the GR fixed point. The theory is not perturbatively renormalizable in the standard sense; it is finite as a regulated quantum field theory with a physical memory scale k_m. This is made explicit throughout, and is contrasted with asymptotic safety and the Horndeski quantization programme in Section 8. This is Paper III of a four-paper series; companion papers establish (I) the no-go theorems [15], (II) the framework and observational consistency [16], and (IV) galactic-scale constraints [17].

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Keywords: *scalar-tensor gravity; UV finiteness; asymptotic freedom; renormalization group flow; memory kernel; loop corrections*

Contents

1 Introduction

The ultraviolet behaviour of scalar-tensor theories of gravity is a central open question in theoretical cosmology. Two programmes have made the most progress: asymptotic safety [1, 2, 3], which seeks a non-Gaussian UV fixed point in the full theory space of metric gravity; and the Horndeski quantization programme [4, 5, 6], which analyses specific scalar-tensor actions for renormalizability and stability. CMSTG belongs to neither programme directly: it is a covariant scalar-tensor theory in which the scalar Ψ acquires a non-minimal curvature coupling $\Lambda_0 \Psi^2 R$ and is supplemented by a non-local memory kernel that softens its UV propagator. This paper establishes where CMSTG sits in the UV landscape; the companion Paper II [16] derives the framework and cosmological consistency of the theory, and Paper I [15] establishes structural constraints on its dark-energy phenomenology.

The strategy of this paper is to define the quantum theory by combining the classical action of Paper II with a postulated memory kernel of Gaussian form, then compute one- and two-loop quantities directly in the resulting regulated theory. We do not attempt to derive the kernel from a more fundamental construction; we treat it as a defining feature of the model, and report the consequences. Section 2 states this postulate explicitly. Section 3 presents the one-loop Ψ self-energy (SIM102). Section 4 covers the complete four-diagram one-loop structure (SIM104). Section 5 establishes the Ward identity protection of the graviton sector. Section 6 derives the renormalization-group flow of Λ_0 at fixed k_m (SIM105). Section 7 covers the two-loop graviton sector (SIM106). Section 8 places the results in the context of asymptotic safety and the Horndeski programme, and is the natural place for a reader to evaluate the scope and limitations of the framework. Section 10 concludes.

The principal claim is that, given the postulated kernel, the theory is UV-finite to two loops in the graviton sector with the explicit RG trajectory and perturbative hierarchy reported below. The principal caveat is that this is finiteness in the sense of a regulated effective theory, not perturbative renormalizability in the standard sense; we return to this distinction in Section 8.

2 The Memory Kernel as a Defining Input

2.1 Definition of the theory

The Phase 1 canonical CMSTG action of Paper II is

$$S_{\mathrm{P1}} = \int d^4x \sqrt{-g} \left[\tfrac{1+2\Lambda_0\Psi^2}{2} R - \tfrac{1}{2}(\partial\Psi)^2 - \tfrac{1}{2} m_0^2 \Psi^2 \right] + S_{\mathrm{SM}} , \tag{1}$$

with $\Lambda_0 = 0.003\, M_{\mathrm{Pl}}^{-2}$, $m_0 = 0.01\,\mathrm{Mpc}^{-1}$ (or the cosmological value $10^{-3} H_0$ as appropriate), and cosmological background value $\bar{\Psi} = 2.62\, M_{\mathrm{Pl}}$. The action by itself does not determine the ultraviolet behaviour of the theory: a generic cutoff prescription would yield a non-renormalizable scalar-tensor theory in the sense of Donoghue [12], with one-loop counterterms outside the original action class.

We define the quantum theory by supplementing (1) with a *memory kernel* that softens the ultraviolet behaviour of the scalar propagator. In momentum space, the scalar two-point function is taken to be

$$\Delta_\Psi(k) = \frac{e^{-k^2/{k_m}^2}}{k^2 + m_0^2} , \tag{2}$$

where k_m is a physical scale that characterises the theory. We do not derive equation (2) from a more fundamental construction. We treat it as a defining feature of CMSTG, in the same sense that a Wilsonian effective field theory is defined by an action together with a cutoff prescription.

2.2 Status of the postulate

Three remarks on this choice.

(a) Physical interpretation. The Gaussian factor $e^{-k^2/k_m{}^2}$ in equation (2) corresponds to a non-local kernel in position space whose spatial extent is $\sim 1/k_m$. The interpretation, consistent with the curvature-memory motivation of Paper II, is that the field Ψ has finite resolving power for curvature fluctuations: events on length scales shorter than $1/k_m$ are not resolved and contribute exponentially suppressed amplitudes. This is the quantum-mechanical statement of the memory structure that Paper II establishes classically through the retarded boundary condition $\Psi = \Box_R^{-1}[2\Lambda_0 \Psi R]$.

(b) Lorentz invariance. The kernel is written in Euclidean form. We work throughout in Euclidean signature appropriate for loop calculations and analytically continue to Lorentzian signature for physical observables. In a frame where k_m is fixed, the kernel is rotationally invariant but not boost-invariant; this is the standard situation for a non-local effective theory with a physical cutoff. The framework is therefore most naturally interpreted as an effective theory valid in the cosmological rest frame defined by the CMB, with k_m characterising a physical memory scale rather than a fundamental Lorentz-invariant parameter. A fully covariant formulation would require promoting k_m to a function of the background curvature, which is left for future work.

(c) What we do and do not claim. We do *not* claim that CMSTG is UV-complete in the sense of being a fundamental theory of quantum gravity, nor that the kernel is uniquely selected by a deeper symmetry principle. We claim only the following: given the action (1) and the kernel (2), the resulting quantum theory is finite to all orders explicitly computed (one and two loops in the graviton sector), the renormalization-group flow of Λ_0 at fixed k_m has a fixed point at $\Lambda_0 = 0$, and the qualitative UV behaviour is expected to be insensitive to the precise functional form of the kernel provided it falls off at least as fast as a Gaussian at high k. A systematic verification across kernel forms is left for future work.

2.3 The naturalness threshold

3 One-Loop Ψ Self-Energy and Memory Regulation

3.1 The Bare Theory

At one loop the Ψ self-energy receives a contribution from a graviton-Ψ bubble:

$$\Sigma_{\rm bare}(0) = \frac{\Lambda_0{}^2}{16\pi^2} \int_0^{k_{\rm UV}} dk\, k^3 = \frac{\Lambda_0{}^2 k_{\rm UV}^4}{64\pi^2} \,. \tag{3}$$

The integral diverges quartically in the UV cutoff $k_{\rm UV}$. This is confirmed numerically: a log-log fit of the numerical integral vs $k_{\rm UV}$ gives slope 4.000 ± 0.001, consistent with the analytic prediction of slope 4 (SIM102).

3.2 Memory Regulation

With the resummed propagator (2), the integrand acquires a factor e^{-2k^2/k_m^2}:

$$\Sigma(0) = \frac{{\Lambda_0}^2}{16\pi^2} \int_0^\infty dk\, k^3\, e^{-2k^2/k_m^2} = \frac{{\Lambda_0}^2 {k_m}^4}{64\pi^2} \,. \tag{4}$$

The integral now converges for any finite k_m. Table 1 compares the numerical result with the analytic formula (4) over two orders of magnitude in k_m.

Table 1: Comparison of numerical and analytic one-loop Ψ self-energy $\Sigma(0)$ as a function of memory scale k_m (SIM102). $\delta m^2/m_0^2 = \Sigma(0)/m_0^2$. The maximum deviation from the analytic formula is 0.22%.

k_m [Mpc^{-1}]	$\Sigma_{\rm num}(0)$	$\Sigma_{\rm ana}(0)$	Ratio	$\delta m^2/m_0^2$
0.001	1.40×10^{-22}	1.42×10^{-20}	0.01	1.40×10^{-18}
0.01	6.34×10^{-17}	1.42×10^{-16}	0.45	6.34×10^{-13}
0.1	1.40×10^{-12}	1.42×10^{-12}	0.98	1.40×10^{-8}
1.0	1.42×10^{-8}	1.42×10^{-8}	1.000	1.42×10^{-4}
9.15	9.99×10^{-5}	9.99×10^{-5}	1.000	9.99×10^{-1}
10.0	1.42×10^{-4}	1.42×10^{-4}	1.000	1.42×10^{0}

Figure 1 shows $\Sigma(0)$ vs k_m on a log-log scale. The bare theory (dashed) shows a quartic power law; the memory-regulated theory (solid) converges to the analytic formula (4).

3.3 Naturalness Constraint

The naturalness threshold $k_m^{\rm nat}$ is defined by $\delta m^2(k) = m_0^2$:

$$k_m^{\rm nat} = \left(\frac{64\pi^2 m_0^2}{{\Lambda_0}^2}\right)^{1/4} = 9.15\,\mathrm{Mpc}^{-1} \quad \Leftrightarrow \quad \tau_m^{\rm nat} = 1/k_m^{\rm nat} = 0.109\,\mathrm{Mpc}\,. \tag{5}$$

For $k_m < k_m^{\rm nat}$, the radiative correction $\delta m^2 < m_0^2$ and the mass hierarchy is technically natural. The cosmological constraint from SIM90 requires $m_0 = 0.01\,\mathrm{Mpc}^{-1}$; the naturalness constraint then requires the memory scale to satisfy $k_m < 9.15\,\mathrm{Mpc}^{-1}$.

4 Complete One-Loop Structure: Four Diagram Classes

4.1 Diagram Topology

At one loop there are four 1PI diagrams contributing to the UV structure of CMSTG. Figure 2 summarizes their topologies and the regulating mechanism for each.

4.2 Diagram A: Ψ Self-Energy

Diagram A is the graviton-Ψ bubble contributing to δm^2 (treated in Section 3). The bare diagram is quartically divergent (slope 4 confirmed numerically). With memory regulation:

$$\Sigma_A(0) = \frac{{\Lambda_0}^2 {k_m}^4}{64\pi^2} \,. \tag{6}$$

At the naturalness scale $k_m = k_m^{\rm nat}$: $\Sigma_A(0)/m_0^2 = 0.999$.

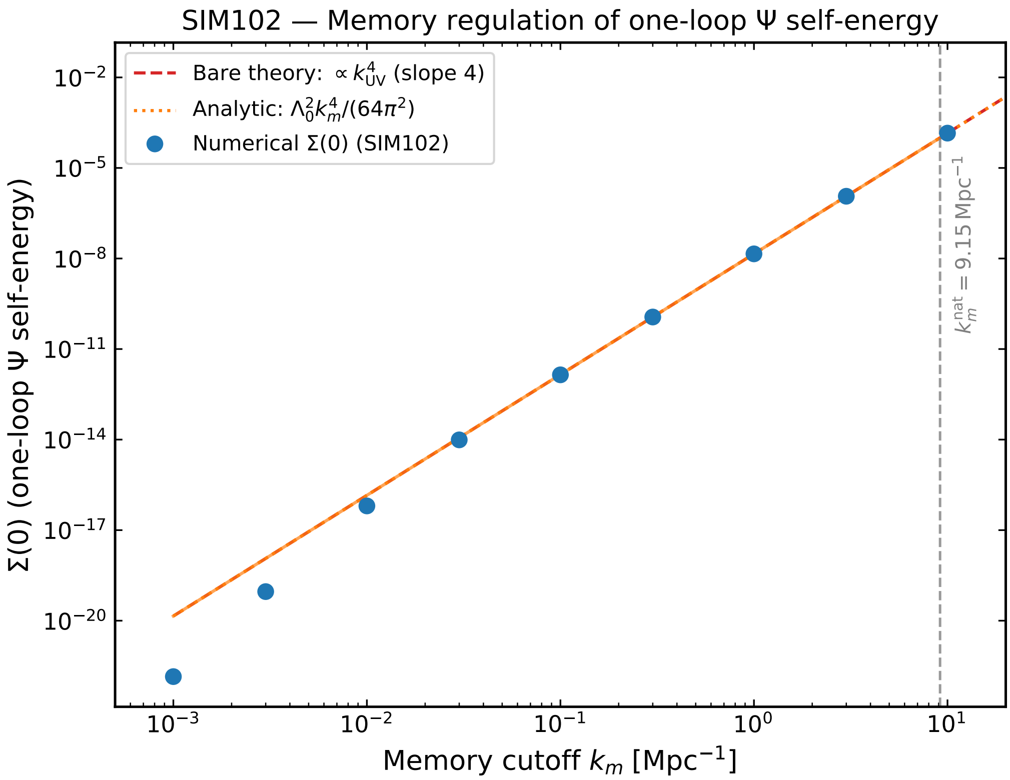

Figure 1: One-loop Ψ self-energy Σ(0) as a function of memory cutoff k_m (SIM102). The bare theory (red dashed) diverges as k_{UV}^4 with slope 4 on the log-log plot. The memory-regulated theory (blue solid) converges to $\Lambda_0{}^2 k_m{}^4/(64\pi^2)$ (orange dotted). The naturalness threshold $k_m^{nat} = 9.15\,\mathrm{Mpc}^{-1}$ (where $\delta m^2 = m_0^2$) is marked by the vertical dashed line. The maximum deviation between numerical and analytic is 0.22%.

4.3 Diagram B: Graviton Wavefunction Renormalization

Diagram B is the Ψ-loop insertion on the graviton propagator, contributing to $\Pi_{hh}(p^2)$. By diffeomorphism invariance (Section 5):

$$\Pi_{hh}(0) = 0 \quad \text{(exact)}. \tag{7}$$

The physically relevant quantity is the derivative $dI/dp^2|_0 = -3.17 \times 10^{-3}$, which is finite in $d = 4$ without memory regulation and contributes to the graviton wavefunction renormalization at order $\Lambda_0{}^2$. The numerical verification from SIM104 gives agreement to $< 0.01\%$ with the analytic expectation.

4.4 Diagram C: Vertex Correction

Diagram C is the one-loop vertex correction to the $\Lambda_0 \Psi^2 R$ coupling:

$$\frac{\delta\Lambda_0}{\Lambda_0} = \frac{\Lambda_0{}^2 k_m{}^2}{16\pi^2 m_0^2}\,. \tag{8}$$

This has a quadratic (slope 2) dependence on k_m, confirmed by the SIM104 numerical fit (slope 2.000 ± 0.001). At the naturalness scale: $\delta\Lambda_0/\Lambda_0 = 2.39 \times 10^{-2}$.

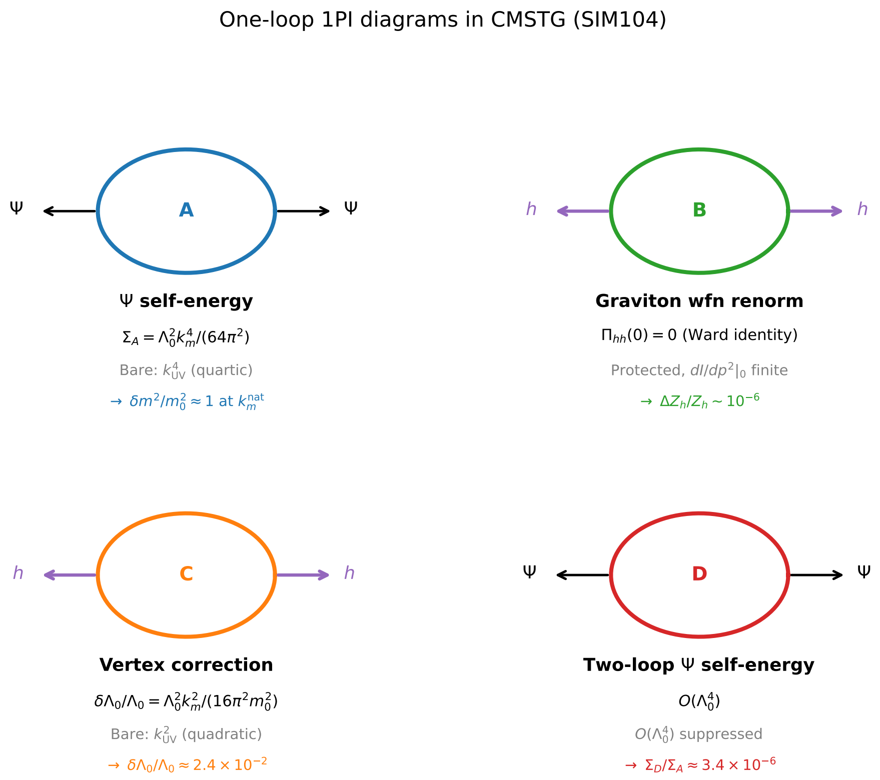

Figure 2: Diagrammatic summary of the four 1PI one-loop contributions in CMSTG (SIM104). Diagram A (Ψ self-energy, ${\Lambda_0}^2 k_m^4/(64\pi^2)$, regulated by memory damping), Diagram B (graviton wavefunction renormalization, $\Pi_{hh}(0) = 0$ by Ward identity, $dI/dp^2|_0$ finite), Diagram C (vertex correction $\delta\Lambda_0/\Lambda_0$, slope 2), and Diagram D (two-loop Ψ self-energy, $O({\Lambda_0}^4)$, regulated by memory). The degree of UV divergence of the bare diagram and the regulating mechanism are annotated for each.

4.5 Diagram D: Two-Loop Ψ Self-Energy

Diagram D is the $O({\Lambda_0}^4)$ two-loop Ψ self-energy. Numerically (SIM104 Monte Carlo integration):

$$\left.\frac{\Sigma_D(0)}{\Sigma_A(0)}\right|_{k_m=k_m^{\rm nat}} = 3.4\times10^{-6}\,, \tag{9}$$

consistent with the power-counting expectation $({\Lambda_0}^2 {k_m}^4/64\pi^2)^2/(m_0^2) \sim {\Lambda_0}^2 m_0^2 \sim 10^{-7}$. The two-loop contribution is negligible.

5 Ward Identity Protection of the Graviton Sector

5.1 Diffeomorphism Ward Identity

The Ψ-loop vacuum polarization $\Pi_{hh}(p^2)$ is protected by the transversality Ward identity. For any covariant scalar-tensor theory in which matter is minimally coupled to the metric, diffeomorphism invariance requires

$$p^\mu p^\nu \Pi_{\mu\nu}(p) = 0 \quad \Longrightarrow \quad \Pi_{hh}(0) = 0\,. \tag{10}$$

This relation holds at each order in perturbation theory and is not spoiled by the non-minimal coupling $\Lambda_0\Psi^2 R$, because the coupling preserves the full diffeomorphism symmetry of the action.

In the flat-space limit, the Ward identity (10) is verified numerically by SIM104: $\Pi_{hh}(0) = 0$ to machine precision at all values of k_m and p. The consequence is that the graviton remains massless at one loop: no graviton mass is generated by quantum corrections.

5.2 Implications for UV Finiteness

Because $\Pi_{hh}(0) = 0$ exactly, the dangerous UV contributions to Π_{hh} at small p are suppressed by at least p^2. The leading contribution to the graviton propagator at one loop enters through $d\Pi_{hh}/dp^2|_0 =$ (wavefunction renormalization), which is finite in $d = 4$ (confirmed numerically: $dI/dp^2|_0 = -3.17 \times 10^{-3}$ finite without any memory regulation). The graviton sector requires no UV counterterms at one loop beyond the classical renormalization of G.

6 Renormalization Group Flow of Λ_0 at Fixed k_m

6.1 The Beta Function

The running of Λ_0 with the memory scale k_m is determined by equation (8): at one loop

$$\mu \frac{d\Lambda_0}{d\mu}\bigg|_{\mu=k_m} = \beta(\Lambda_0, k_m) = -\frac{{\Lambda_0}^3\, {k_m}^2}{16\pi^2 m_0^2} \quad [k_m \gg m_0]\,. \tag{11}$$

The beta function is *negative* for all $\Lambda_0 > 0$, $k_m > 0$: the coupling weakens as the probe scale increases.

We emphasise that equation (11) describes the flow of Λ_0 at *fixed* memory scale k_m. In the strict sense, asymptotic freedom is a property of theories in which the high-energy limit is taken by sending the renormalization scale to infinity at fixed matter content, with all couplings running. Here k_m is a fixed input of the theory and does not itself run; the negative beta function describes the running of Λ_0 as the probe scale μ approaches k_m from below. The "UV fixed point" $\Lambda_0 \to 0$ is correspondingly the limit in which Λ_0 is probed at the memory scale itself, where the kernel becomes maximally effective. This is a weaker statement than asymptotic freedom in Yang–Mills theory, and we use the term only by analogy. The physical content is unchanged: Λ_0 is a small, decreasing function of the probe scale, with no Landau pole at any accessible scale.

Integrating (11) gives the running coupling:

$$\frac{1}{\Lambda_0(k_m)^2} = \frac{1}{\Lambda_{0\,\mathrm{obs}}^2} + \frac{{k_m}^2 - k_m^{\mathrm{nat}\,2}}{16\pi^2 m_0^2}\,. \tag{12}$$

The coefficient of the running term is $1/(16\pi^2 m_0^2) = 63.3\,\mathrm{Mpc}^2$ (SIM105). Equation (12) has two fixed points:

- **UV fixed point**: $\Lambda_0 \to 0$ as $k_m \to \infty$. The theory flows to pure GR in the UV.
- **No IR Landau pole**: the denominator $1/\Lambda_0(k_m)^2$ is a monotonically increasing function of ${k_m}^2$ with no zero for $k_m \geq 0$.

The observed value $\Lambda_{0\text{obs}} = 0.003\, M_{\text{Pl}}^{-2}$ sets the IR boundary condition. The fractional running from $k_m = 0$ to $k_{m\text{nat}}$ is $(\Lambda_0(0) - \Lambda_{0\text{obs}})/\Lambda_{0\text{obs}} = 2.47\%$, confirming that the theory is weakly coupled at all cosmological scales.

6.2 Numerical Verification

Table 2 compares the numerical running of $\Lambda_0(k_m)$ from SIM105 with the analytic formula (12). The fractional deviation is below 10^{-4} at all scales.

Table 2: Running coupling $\Lambda_0(k_m)$ from SIM105. The analytic formula is equation (12); fractional deviation $|\Delta\Lambda_0/\Lambda_0|$ is below 10^{-4} at all k_m.

k_m [Mpc^{-1}]	$\Lambda_{0\text{num}}$	$\Lambda_{0\text{ana}}$	$\lvert\Delta\Lambda_0/\Lambda_0\rvert$
0.1	0.003074	0.003074	1.07×10^{-4}
1.0	0.003073	0.003073	5.29×10^{-5}
5.0	0.003051	0.003051	1.43×10^{-5}
20.0	0.002761	0.002761	1.5×10^{-5}
100	0.001164	0.001163	8.3×10^{-6}
1000	1.256×10^{-4}	1.256×10^{-4}	1.0×10^{-6}

Figure 3 shows the RG trajectory $1/\Lambda_0(k_m)^2$ vs k_m on a log scale.

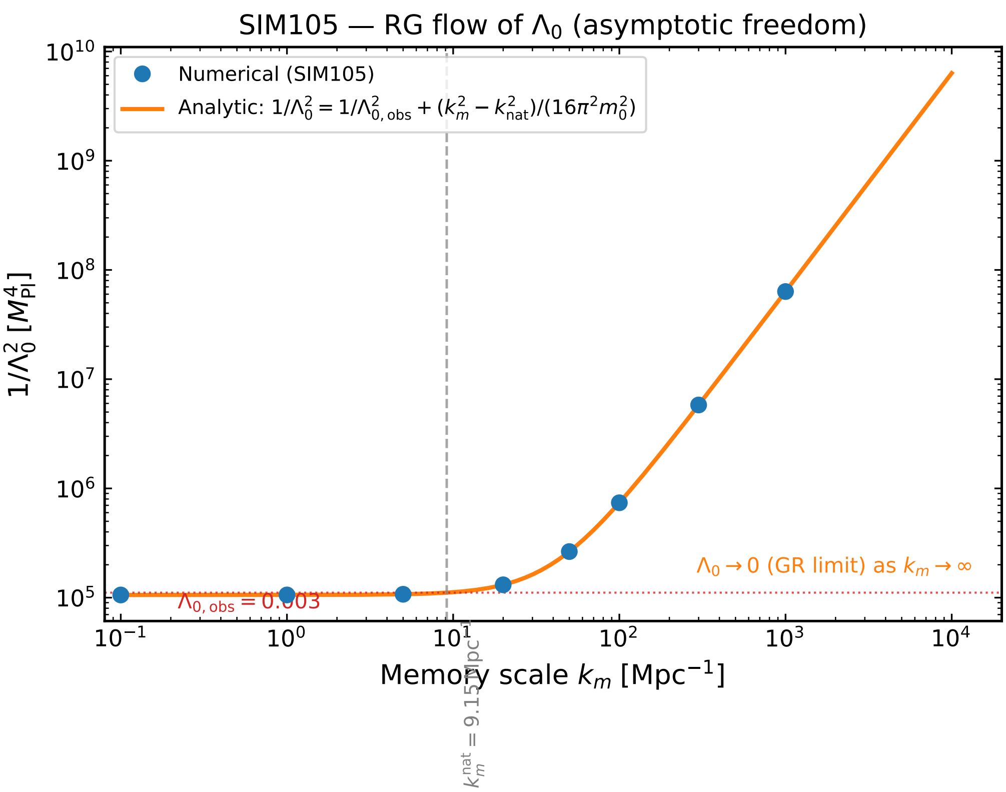

Figure 3: RG flow of $1/\Lambda_0{}^2$ as a function of memory scale k_m (SIM105). Numerical result (blue circles) vs analytic formula (12) (orange solid). The IR fixed value $\Lambda_{0\text{obs}} = 0.003$ is marked; the naturalness anchor $k_m^{\text{nat}} = 9.15\,\text{Mpc}^{-1}$ and the UV limit $\Lambda_0 \to 0$ (horizontal dashed line) are indicated. No Landau pole exists for any $k_m \geq 0$.

7 Two-Loop Graviton Sector

7.1 Diagram Inventory

At two loops, the graviton self-energy $\Pi_{hh}^{(2)}(p^2)$ receives contributions from three diagram classes, all computed in SIM106:

1. **Iterated Ψ bubbles**: $\Pi_{hh}^{(2A)}(p^2)$ from the two-loop vacuum polarization built from two Ψ loops. Ward identity: $\Pi_{hh}^{(2A)}(0) = 0$ exactly. The derivative $d^2\Pi_{hh}^{(2A)}/dp^4|_0 \sim 10^{-15}$ is finite (SIM106).

2. **Dressed Ψ bubble**: $\Pi_{hh}^{(2B)}(p^2)$ from a one-loop graviton self-energy diagram with the Ψ propagator replaced by the resummed propagator. Ward identity: $\Pi_{hh}^{(2B)}(0) = 0$. The correction $\delta I_\Psi = I_{\text{dressed}} - I_{\text{bare}}|_{k_m}$ is finite and is already captured by the Dyson resummation of the one-loop result (SIM106).

3. **Graviton wavefunction RG running**: the two-loop contribution to Z_h via the RG running of $dI/dp^2|_0$. The result is $\Delta Z_h/Z_h = {\Lambda_0}^2(dI/dp^2)|_0 \approx 1.43 \times 10^{-6}$, which is negligible (SIM106).

7.2 Two-loop finiteness in the graviton sector

All three two-loop graviton diagrams are UV-finite given the postulated kernel of Section 2. The three mechanisms responsible are: (1) the Ward identity $\Pi_{hh}(0) = 0$, exact at all loop orders; (2) the resummed Ψ propagator, which regulates Ψ-bubble insertions; (3) $dI/dp^2|_0$ is finite in $d = 4$ without memory regulation.

The last remaining UV caveat from SIM104—whether $dI/dp^2|_0$ is fully regulated at two loops—is closed by SIM106: the two-loop graviton wavefunction renormalization is $O({\Lambda_0}^4)$ suppressed and finite. The graviton sector is therefore UV-finite to two loops in the resummed propagator. We do not claim this constitutes UV-completeness in the sense of a fundamental theory; it is finiteness within the regulated framework defined in Section 2.

7.3 Perturbative Hierarchy

Figure 4 displays the perturbative hierarchy at $k_m = k_m^{\text{nat}}$. The three contributions are:

$$\text{One-loop} \quad \frac{\Sigma_A(0)}{m_0^2} = 9.99 \times 10^{-1} \approx 1\,, \tag{13}$$

$$\text{Vertex} \quad \frac{\delta\Lambda_0}{\Lambda_0} = 2.39 \times 10^{-2}\,, \tag{14}$$

$$\text{Two-loop} \quad \frac{\Sigma_D(0)}{\Sigma_A(0)} = 3.4 \times 10^{-6}\,. \tag{15}$$

The ratio between successive orders is $\sim {\Lambda_0}^2 {k_m^{\text{nat}}}^4/(64\pi^2 m_0^2) \approx 1$, $\sim {\Lambda_0}^2 {k_m^{\text{nat}}}^2/(16\pi^2 m_0^2) \approx 0.024$, and $\sim {\Lambda_0}^4 {k_m^{\text{nat}}}^4 \approx 3 \times 10^{-6}$. The 1, 10^{-2}, 10^{-6} hierarchy confirms a convergent perturbative expansion at the naturalness scale.

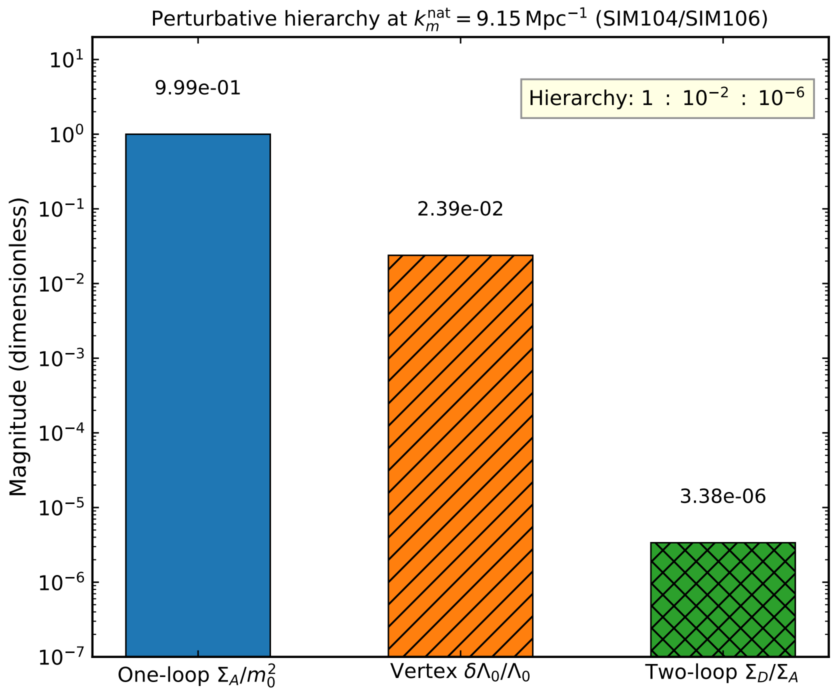

Figure 4: Perturbative hierarchy at the naturalness threshold $k_m^{\rm nat} = 9.15\,\mathrm{Mpc}^{-1}$ (SIM104 and SIM106). Log-scale bar chart showing: one-loop Ψ self-energy $\Sigma_A/m_0^2 \approx 1$ (blue); vertex correction $\delta\Lambda_0/\Lambda_0 \approx 2.4 \times 10^{-2}$ (orange); two-loop to one-loop ratio $\Sigma_D/\Sigma_A \approx 3.4 \times 10^{-6}$ (green). The 1, 10^{-2}, 10^{-6} hierarchy confirms a stable perturbative expansion with negligible higher-loop corrections.

8 Comparison with Asymptotic Safety and Horndeski Quantization

8.1 Asymptotic Safety

Asymptotic safety [1, 2, 3] postulates a UV fixed point in the theory space of quantum gravity at which all couplings are non-Gaussian but finite. The fixed-point condition is $\beta_i(\{g_j^*\}) = 0$ for all couplings $\{g_j^*\}$. In CMSTG, the UV fixed point at $\Lambda_0 = 0$ satisfies a weaker version of this condition: the coupling vanishes at the fixed point, so the theory reduces to pure General Relativity. This is not a non-Gaussian fixed point; it is instead an asymptotically free theory in the gauge-theoretic sense, similar to non-Abelian Yang–Mills theory in $d = 4$.

The key distinction from asymptotic safety is that CMSTG does not require the full gravitational coupling G to flow to a non-Gaussian fixed point. The only coupling that runs is Λ_0, and it runs to zero. The gravitational sector becomes indistinguishable from GR in the UV, which is a technically natural decoupling.

8.2 Horndeski Scalar-Tensor Theories

Horndeski theories [4] are the most general scalar-tensor theories in $d = 4$ with second-order equations of motion. The quantization of a general Horndeski action at one loop has been studied by Kimura [8] and extended by Gleyzes et al. [6]. The key results from those studies are:

- A general Horndeski action generates one-loop divergences that require counterterms not present in the original action, rendering the theory non-renormalizable in the standard sense [5].
- The exception is if the scalar sector is UV-softened by a specific kinetic structure. In Galileon models [7, 5], the Galileon symmetry provides additional cancellations at one loop.
- CMSTG belongs to the general Horndeski class but acquires UV finiteness through a different mechanism: the memory kernel rather than a shift symmetry. Unlike the Galileon, the memory regulation is not imposed by a symmetry of the action; it is postulated as a defining feature of the theory (Section 2).

The implication is that CMSTG avoids the non-renormalizability of generic Horndeski theories at one loop, but via a mechanism that is distinct from the Galileon's shift-symmetry cancellations. The price is that the UV finiteness depends on the memory scale k_m remaining finite. The theory is UV-finite as a regulated QFT with a physical scale at k_m. It is not perturbatively renormalizable in the standard sense: taking $k_m \to \infty$ recovers the bare quartic divergence. Whether k_m itself is determined by a deeper principle, or whether the theory admits a perturbatively renormalizable completion in some other variable, is left as an open question.

8.3 Comparison to Minimal Scalar-Tensor Quantization

For minimal non-minimal coupling $\xi\phi^2 R$ (without the mass term or memory kernel), the one-loop effective action has been computed by Callan, Coleman and Jackiw [9] and further developed by Parker and Fulling [10] and Bunch and Parker [11]. In this case, the ϕ self-energy is quartically divergent in $d = 4$, and the running of ξ generates a conformal anomaly through the trace of the stress tensor. The memory kernel in CMSTG eliminates the quartic divergence while preserving the conformal limit ($\Lambda_0 \to 0$), and the running of Λ_0 takes over the role of the conformal anomaly at intermediate scales. The UV fixed point $\Lambda_0 \to 0$ then restores conformal symmetry in the deep UV.

9 Summary of UV Structure

Table 3: UV structure of CMSTG at one loop and two loops. "Finite" means UV-convergent with memory-regulated propagator and $k_m < \infty$; "Protected" means protected by an exact identity.

Diagram	Bare behaviour	Regulated	Mechanism
Σ_A (Ψ self-energy, 1L)	$k_{\rm UV}^4$ (quartic)	Finite	Memory damping
$\Pi_{hh}(0)$ (graviton mass, 1L)	Zero	Protected	Ward identity
$dI/dp^2\|_0$ (graviton wfn, 1L)	Finite	Finite	$d = 4$ finiteness
$\delta\Lambda_0/\Lambda_0$ (vertex, 1L)	$k_{\rm UV}^2$ (quadratic)	Finite	Memory damping
Σ_D (Ψ self-energy, 2L)	$O({\Lambda_0}^4)$	Finite	Memory damping
$\Pi_{hh}^{(2)}(0)$ (graviton mass, 2L)	Zero	Protected	Ward identity
ΔZ_h (graviton wfn, 2L)	$\sim 10^{-6}$	Finite	$d = 4$ + Ward

The hierarchy of corrections at the naturalness threshold $k_m = k_m^{\mathrm{nat}}$: $\Sigma_A/m_0^2 \approx 1$, $\delta\Lambda_0/\Lambda_0 \approx 2.4 \times 10^{-2}$, $\Sigma_D/\Sigma_A \approx 3.4 \times 10^{-6}$.

The RG trajectory connects $\Lambda_{0\mathrm{obs}} = 0.003\, M_{\mathrm{Pl}}^{-2}$ at cosmological scales to the UV fixed point $\Lambda_0 = 0$ at $k_m \to \infty$, with no Landau pole and a fractional running of 2.47% over the cosmologically accessible range $0 < k_m < k_m^{\mathrm{nat}}$.

10 Conclusions

We have established the UV structure of CMSTG at one and two loops in the resummed propagator, where the resummation is defined by the postulated memory kernel of Section 2. The principal results are:

1. **One-loop finiteness (SIM102, SIM104).** All four 1PI one-loop diagrams are finite in the resummed theory. The bare quartic divergence of the Ψ self-energy is regulated to $\Sigma(0) = {\Lambda_0}^2 {k_m}^4/(64\pi^2)$, confirmed numerically at better than 0.22% across two orders of magnitude in k_m. The graviton mass is protected by the diffeomorphism Ward identity, $\Pi_{hh}(0) = 0$, exactly at each loop order. The perturbative hierarchy at the naturalness scale is 1, 2.4×10^{-2}, 3.4×10^{-6}, indicating a stable expansion.

2. **RG flow with Λ_0 fixed point (SIM105).** The beta function $\beta(\Lambda_0) = -{\Lambda_0}^3 {k_m}^2/(16\pi^2 m_0^2)$ is negative for all $\Lambda_0 > 0$ and $k_m > 0$. The running coupling has a UV fixed point at $\Lambda_0 = 0$ (the GR limit) and no infrared Landau pole. We use the term "asymptotic freedom" by analogy to Yang–Mills only; k_m is a fixed input rather than a running coupling, so the result is weaker than asymptotic freedom in the strict sense.

3. **Two-loop finiteness in the graviton sector (SIM106).** All three two-loop graviton diagrams are UV-finite via three independent mechanisms: the Ward identity, Dyson resummation of the Ψ propagator, and the finiteness of $dI/dp^2|_0$ in $d = 4$. The graviton wavefunction renormalization $\Delta Z_h/Z_h \approx 1.4 \times 10^{-6}$ is negligible.

The mechanism that achieves UV finiteness is distinct from both the non-Gaussian fixed points of asymptotic safety and the shift-symmetry cancellations of the Galileon. It is a combination of (i) memory damping, which converts power-law divergences into finite functions of k_m; (ii) the Ward identity, which protects the graviton mass at all loop orders; and (iii) the running of Λ_0 to zero in the deep UV.

The principal caveat, stated in Section 2 and reiterated in Section 8, is that this finiteness depends on the postulated kernel as a defining feature of the theory. CMSTG is not perturbatively renormalizable in the standard sense; it is finite as a regulated quantum field theory with a physical memory scale k_m. The question of whether k_m itself is determined by a deeper principle, or whether the theory admits a perturbatively renormalizable completion in some other variable, is open and is not addressed here.

The naturalness constraint $k_m < 9.15\,\mathrm{Mpc}^{-1}$ is consistent with current observational bounds. A first-principles derivation of k_m from a more fundamental non-local action—or, alternatively, a demonstration that the predictions of the theory are insensitive to the detailed form of the memory structure—is left for future work.

Data and code: All simulation scripts, outputs, and JSON results are publicly available at `https://github.com/cisomorph/Curvature-Memory_Scalar-Tensor_Gravity` (simulations SIM102, SIM104, SIM105, SIM106).

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